Table 1. Comparative Analysis of our proposed approach against SRCNN, EDSR, RCAN and SwinIR in terms of model parameters, size in MB, MACS/FLOPS, Inference Time and Memory Consumption during inference.

Dataset	Scale	Model	Params	Model Size (in MB)	MACS/FLOPS	Inference Time	Memory Consumption
Test	x2	SRCNN	20,099	0.08 MB	0.0819 G / 0.1638 G	0.6385 ms/image	2.87 MB
		EDSR	1,369,883	5.23 MB	1.406 G / 2.811 G	4.3955 ms/image	138.86 MB
		RCAN	4,094,603	15.62 MB	4.165 G / 8.329 G	18.6847 ms/image	198.20 MB
		SwinIR	3,145,212	12.00 MB	3.204 G / 6.407 G	41.9210 ms/image	164.90 MB
		Ours	156,892	0.60 MB	0.085 G / 0.169 G	13.6210 ms/image	152.84 MB
GBM	x4	SRCNN	20,099	0.08 MB	0.082 G / 0.164 G	0.6541 ms/image	34.84 MB
		EDSR	1,517,595	5.79 MB	0.508 G / 1.015 G	4.6091 ms/image	52.81 MB
		RCAN	4,537,739	17.31 MB	1.160 G / 2.319 G	18.5381 ms/image	86.18 MB
		SwinIR	3,176,352	12.12 MB	0.809 G / 1.618 G	34.2188 ms/image	85.25 MB
		Ours	190,236	0.73 MB	0.067 G / 0.135 G	12.5803 ms/image	31.18 MB

Table 2. Ablation Analysis performed on x2 scale highlighting the importance and contribution of FastKAN in the architecture.

Scale	Data	W/o FastKAN	OURS
x2	Test	42.2963 / 1.0000	42.7046 / 1.0000
	Val1	44.5416 / 0.9830	44.9817 / 0.9863
	Val2	45.3236 / 0.9836	45.5535 / 0.9845
	GBM	38.3221 / 0.9583	38.5483 / 0.9602
	PRAD	38.5111 / 0.9772	38.8635 / 0.9793
	LUAD	35.7677 / 0.9384	35.9167 / 0.9395

Table 3. Gradient Magnitude Similarity Deviation (GSMD) full-reference image quality metric performance evaluation on x2 scale Test and GBM datasets indicating better image reconstruction by our proposed approach.

Dataset	Scale	Model	Params	GSMD Score
Test	x2	SRCNN	20,099	0.062705
		EDSR	1,369,883	0.061633
		RCAN	4,094,603	0.057851
		SwinIR	3,145,212	0.055722
		Ours	156,892	0.053781
GBM	x2	SRCNN	20,099	0.095388
		EDSR	1,369,883	0.093534
		RCAN	4,094,603	0.093821
		SwinIR	3,145,212	0.088634
		Ours	156,892	0.085368

Table 4. Details of sizes of the datasets in terms of gigabytes consumed for storage, indicating the amount of storage reduction with ISR and downsampling technique.

Dataset	2048*2048 Size (GB)	1024*1024 Size (GB)	512*512 Size (GB)
Test	11.62	2.66	0.72
Val1	6.36	1.46	0.4
Val2	6.05	1.3875	0.37